

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 5: Due by 2:45pm on Tuesday February 27, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [45 points] Consider the 2nd order boundary value problem $y''(t) = -y(t) + 2\cos(t)$ with the boundary constraints: $y(0) = 0$ and $y(\pi/2) = \pi/2$.

- (A). [5 points] Prove that $y(t) = t \sin(t)$ solves the BVP.
- (B). [15 points] Solve the BVP using the shooting method with $N = 20$ time steps. (Use the forward Euler method.) Plot the approximations y_i versus t_i in one plot, and in a second plot show the errors $e_i = |y(t_i) - y_i|$ versus t_i . Record the maximum error $E_\infty = \max_i e_i$.
- (C). [10 points] Repeat Problem 1B, but implement the shooting method using different choices for the number of time steps, e.g., $N = 200$ and $N = 2000$. How does the maximum error vary with N ?
- (D). [15 points] Repeat Problem 1C (i.e., using $N = 200$ and $N = 2000$) but this time implement the shooting method using the explicit trapezoidal method. Now how does the maximum error vary with N ?

2. [20 points]

- (A). [5 points] Prove that $y'(t) = \frac{y(t) - y(t-h)}{h} + \mathcal{O}(h)$ (backward difference)
- (B). [5 points] Prove that $y'(t) = \frac{y(t+h) - y(t-h)}{2h} + \mathcal{O}(h^2)$ (centered difference).
Hint: Taylor expand both $y(t+h)$ and $y(t-h)$, subtract these 2 equations, and solve for $y'(t)$.
- (C). [10 points] Let $y(t) = e^t$ and implement a numerical experiment to validate that the backward and forward differences are $\mathcal{O}(h)$, whereas the centered difference is $\mathcal{O}(h^2)$.

3. [35 points] Consider the 2nd order boundary value problem $y''(t) = y(t) + \frac{2}{3}e^t$ with the boundary constraints: $y(0) = 0$ and $y(1) = \frac{1}{3}e^1$.

- (A). [5 points] Prove that $y(t) = te^t/3$ solves the BVP.
- (B). [5 points] Let $N = 4$ (i.e., $h = 1/4$) and write down the associated finite difference equations for y_0, y_1, y_2, y_3 and y_4 .
- (C). [10 points] Use a linear solver to obtain these y_i values, and make two plots. The first should compare the estimates y_i to the true solution, and the second should plot the approximation error versus t_i .
- (D). [15 points] Repeat Problem 3C two times for the two choices $N = 40$ and $N = 400$. Describe how the error changes as N increases.